LAB 2

NETWORK COMMANDS FOR TESTING & TROUBLESHOOTING

OBJECTIVES

- To explore and understand commonly used network commands for testing and troubleshooting connectivity issues.
- To observe the working and analyse the output of basic diagnostic commands in the Windows operating system.
- To develop practical skills in identifying and resolving common network problems using command-line tools.
- To gain hands-on experience in verifying network configurations and ensuring reliable communication between devices.

THEORY :

Network commands are important tools used for testing, monitoring, and troubleshooting computer networks. They help both administrators and users detect connectivity problems, check IP configurations, identify routing errors, and resolve communication failures. In the Windows operating system, these commands are typically executed through the Command Prompt (CMD), providing quick and reliable diagnostic information.

Some of the commonly used commands include ping for checking connectivity, tracert for tracing the route of packets, and ipconfig for viewing IP configuration details. Other useful commands are nslookup for DNS queries, netstat -a for monitoring active connections, and pathping for analyzing network paths.

Additionally, commands like route, arp -a, hostname, getmac, and nbtstat provide deeper insights into routing tables, MAC addresses, and system identification, making them essential for effective troubleshooting and network management.

COMMANDS

Here, all the commands are written according to Windows Operating System's Command prompt whose syntax may vary according to different Operating System. However, the working or functioning of the command is almost same for different Operating System too. It differs syntactically not functionally.

The various commands that are discussed here with the full command box showing the output of each network command are such as ping, tracert, ipconfig, netstat, nslookup, route, nbtstat, etc.

1. Ping

Used to test connectivity between your computer and another host by sending ICMP echo requests.

Syntax: **ping <hostname or IP address>**

```
C:\Users\suraj>ping www.google.com

Pinging www.google.com [2404:6800:4002:829::2004] with 32 bytes of data:
Reply from 2404:6800:4002:829::2004: time=29ms
Reply from 2404:6800:4002:829::2004: time=32ms
Reply from 2404:6800:4002:829::2004: time=30ms
Reply from 2404:6800:4002:829::2004: time=31ms

Ping statistics for 2404:6800:4002:829::2004:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 29ms, Maximum = 32ms, Average = 30ms
```

2. Tracert

Traces the route packets take to reach a destination, showing each hop along the path.

Syntax: **tracert <hostname or IP address>**

```
C:\Users\suraj>tracert www.google.com

Tracing route to www.google.com [2404:6800:4002:829::2004]
over a maximum of 30 hops:

  1     8 ms     3 ms     2 ms  2400-1A00-B060.ip6.wlink.com.np [2400:1a00:b060:4f3d::1]
  2    21 ms    11 ms     8 ms  2400-1a00-b1a6.ip6.wlink.com.np [2400:1a00:b1a6::1]
  3     3 ms     6 ms     5 ms  2400:1a00:0:1::234
  4    20 ms    12 ms     5 ms  2400:1a00:0:40::50
  5     *        7 ms     *     2400:1a00:0:42::121
  6    18 ms    13 ms    20 ms  2400:1a00:0:41::170
  7    15 ms    10 ms     8 ms  2400:1a00:0:41::128
  8    11 ms     6 ms     6 ms  2400:1a00:dccc:1:72:9:128:67
  9     *        *        *     Request timed out.
 10    32 ms    43 ms    33 ms  2001:4860:1:1::126a
 11    34 ms    26 ms    28 ms  2001:4860:0:1::78d3
 12    40 ms    33 ms    38 ms  2001:4860:0:1::53a3
 13    34 ms    31 ms    32 ms  tzdelb-at-in-x04.1e100.net [2404:6800:4002:829::2004]

Trace complete.
```

3. Ipconfig

Displays IP configuration details such as IP address, subnet mask, and default gateway.

Syntax: **ipconfig**
ipconfig /all

```
C:\Users\suraj>ipconfig

Windows IP Configuration


Unknown adapter McAfee VPN:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : worldlink.com.np
   IPv6 Address. . . . . . . . . . . : 2400:1a00:b060:4f3d:f2f6:60c3:ae81:988d
   Temporary IPv6 Address. . . . . . : 2400:1a00:b060:4f3d:5c30:b074:e52a:b022
   Link-local IPv6 Address . . . . . : fe80::fafb:2f02:5555:3727%12
   IPv4 Address. . . . . . . . . . . : 192.168.1.107
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%12
                                       192.168.1.254

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

4. Nslookup

Queries DNS servers to obtain domain name or IP address mapping.

Syntax: **nslookup <hostname>**

```
C:\Users\suraj>nslookup www.google.com
Server:  vip6-safenet-kmd01.wlink.com.n
Address:  2400:1a00:0:32::165

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4002:809::2004
          142.250.193.132
```

5. Netstat -a

Shows active TCP/UDP connections and listening ports on the system.

Syntax: **netstat -a**

```
C:\Users\suraj>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            EXPERTs-BOOK:0         LISTENING
  TCP    0.0.0.0:445            EXPERTs-BOOK:0         LISTENING
  TCP    0.0.0.0:3306           EXPERTs-BOOK:0         LISTENING
  TCP    0.0.0.0:5040           EXPERTs-BOOK:0         LISTENING
  TCP    0.0.0.0:7680           EXPERTs-BOOK:0         LISTENING
  TCP    0.0.0.0:33060          EXPERTs-BOOK:0         LISTENING
  TCP    0.0.0.0:49688          EXPERTs-BOOK:0         LISTENING
  TCP    127.0.0.1:24830        EXPERTs-BOOK:0         LISTENING
  TCP    127.0.0.1:27017        EXPERTs-BOOK:0         LISTENING
  TCP    127.0.0.1:45112        EXPERTs-BOOK:0         LISTENING
  TCP    127.0.0.1:49350        EXPERTs-BOOK:0         LISTENING
  TCP    127.0.0.1:49351        EXPERTs-BOOK:0         LISTENING
  TCP    127.0.0.1:49669        EXPERTs-BOOK:49670     ESTABLISHED
  TCP    127.0.0.1:49670        EXPERTs-BOOK:49669     ESTABLISHED
  TCP    127.0.0.1:49671        EXPERTs-BOOK:49672     ESTABLISHED
  TCP    127.0.0.1:49672        EXPERTs-BOOK:49671     ESTABLISHED
  TCP    127.0.0.1:51382        EXPERTs-BOOK:49350     TIME_WAIT
```

6. Pathping

Combines ping and tracert to test connectivity and analyze packet loss across each hop.

Syntax: **pathping <hostname or IP address>**

```
C:\Users\suraj>pathping www.google.com

Tracing route to www.google.com [2404:6800:4002:808::2004]
over a maximum of 30 hops:
  0  EXPERTs-BOOK.worldlink.com.np [2400:1a00:b060:4f3d:5c30:b074:e52a:b022]
  1  2400-1A00-B060.ip6.wlink.com.np [2400:1a00:b060:4f3d::1]
  2  2400-1a00-b1a6.ip6.wlink.com.np [2400:1a00:b1a6::1]
  3  2400:1a00:0:1::234
  4  2400:1a00:0:40::50
  5  2400:1a00:0:42::121
  6  2400:1a00:0:41::170
  7  2400:1a00:0:41::128
  8  2400:1a00:dccc:1:72:9:128:67
  9     *        *        *
Computing statistics for 200 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           EXPERTs-BOOK.worldlink.com.np [2400:1a00:b060:4f3d:5c30:b074:e52a:b022]
                                0/ 100 =  0%   |
  1    2ms     0/ 100 =  0%     0/ 100 =  0%  2400-1A00-B060.ip6.wlink.com.np [2400:1a00:b060:4f3d::1]
                                0/ 100 =  0%   |
  2   11ms     0/ 100 =  0%     0/ 100 =  0%  2400-1a00-b1a6.ip6.wlink.com.np [2400:1a00:b1a6::1]
                                0/ 100 =  0%   |
  3    9ms     0/ 100 =  0%     0/ 100 =  0%  2400:1a00:0:1::234
                                0/ 100 =  0%   |
  4    9ms     0/ 100 =  0%     0/ 100 =  0%  2400:1a00:0:40::50
                                0/ 100 =  0%   |
  5    9ms     0/ 100 =  0%     0/ 100 =  0%  2400:1a00:0:42::121
                                0/ 100 =  0%   |
  6   11ms     0/ 100 =  0%     0/ 100 =  0%  2400:1a00:0:41::170
                                0/ 100 =  0%   |
  7   13ms     0/ 100 =  0%     0/ 100 =  0%  2400:1a00:0:41::128
                                1/ 100 =  1%   |
  8   12ms     1/ 100 =  1%     0/ 100 =  0%  2400:1a00:dccc:1:72:9:128:67

Trace complete.
```

7. Route

Displays and modifies the routing table used by the system.

Syntax: **route print**

route add <destination> mask <subnet mask> <gateway>

```
C:\Users\suraj>route print
===========================================================================
Interface List
  7...00 ff 42 fb 9d d1 ......TAP-Windows Adapter V9
 18...28 95 29 86 47 15 ......Microsoft Wi-Fi Direct Virtual Adapter
  9...2a 95 29 86 47 14 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 12...28 95 29 86 47 14 ......Intel(R) Wi-Fi 6E AX211 160MHz
 17...28 95 29 86 47 18 ......Bluetooth Device (Personal Area Network)
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    192.168.1.254    192.168.1.107     45
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.1.0    255.255.255.0         On-link     192.168.1.107    301
    192.168.1.107  255.255.255.255         On-link     192.168.1.107    301
    192.168.1.255  255.255.255.255         On-link     192.168.1.107    301
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     192.168.1.107    301
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     192.168.1.107    301
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
 12   4141 ::/0                     fe80::1
  1    331 ::1/128                  On-link
 12   4141 2400:1a00:b060:4f3d::/64 On-link
 12     61 2400:1a00:b060:4f3d::/64 fe80::1
 12    301 2400:1a00:b060:4f3d:5c30:b074:e52a:b022/128
                                    On-link
 12    301 2400:1a00:b060:4f3d:f2f6:60c3:ae81:988d/128
                                    On-link
 12    301 fe80::/64                On-link
 12    301 fe80::fafb:2f02:5555:3727/128
                                    On-link
  1    331 ff00::/8                 On-link
 12    301 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

8. Arp -a

Shows the Address Resolution Protocol (ARP) table, mapping IP addresses to MAC addresses.

Syntax: **arp -a**

```
C:\Users\suraj>arp -a

Interface: 192.168.1.107 --- 0xc
  Internet Address      Physical Address      Type
  192.168.1.254         c4-48-fa-20-63-f0     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

9. Hostname

Displays the name of the computer (host) on the network.

Syntax: **hostname**

```
C:\Users\suraj>hostname
EXPERTs-BOOK
```

10. Getmac

Displays the MAC address of the network adapter(s) on the system.

Syntax: **getmac**

```
C:\Users\suraj>getmac

Physical Address    Transport Name
=================== ============================================
28-95-29-86-47-14   \Device\Tcpip_{6A8C178F-0FC4-4FA0-A14D-A[redacted]
28-95-29-86-47-18   Media disconnected
00-FF-42-FB-9D-D1   Media disconnected
```

11. Nbtstat

Displays NetBIOS over TCP/IP information, including name tables and sessions.

Syntax: **nbtstat -n**
nbtstat -s

```
C:\Users\suraj> nbtstat -n

McAfee VPN:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Wi-Fi:
Node IpAddress: [192.168.1.107] Scope Id: []

                NetBIOS Local Name Table

       Name               Type         Status
    ---------------------------------------------
    EXPERTS-BOOK   <20>  UNIQUE      Registered
    EXPERTS-BOOK   <00>  UNIQUE      Registered
    WORKGROUP      <00>  GROUP       Registered

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Local Area Connection* 1:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Local Area Connection* 2:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache
```

DISCUSSION

In this lab, we practiced using different network commands in the Windows operating system to test and troubleshoot connectivity. Commands such as ping, tracert, and ipconfig helped us verify communication between devices and check IP configurations, while tools like nslookup, netstat, and pathping provided deeper insights into DNS resolution, active connections, and packet loss. By observing the outputs of these commands, we understood how each one contributes to diagnosing specific issues such as unreachable hosts, incorrect configurations, or routing errors. This hands-on practice reinforced theoretical knowledge and highlighted the importance of command-line utilities in real-world network troubleshooting.

CONCLUSION

The lab successfully demonstrated the role of basic network commands in identifying and resolving connectivity problems. Through practical exposure, we gained confidence in using diagnostic tools to monitor connections, verify configurations, and troubleshoot common issues. Overall, the activity strengthened our understanding of network operations and prepared us to apply these skills in future scenarios where quick and accurate problem-solving is essential.